

DM-2026-006

SendCUIEmail Development Team
Software Engineering

DATE: January 21, 2026

TO: Project Record

SUBJECT: Beta Testing Readiness Assessment

OFFICE OF PRIMARY RESPONSIBILITY: SendCUIEmail Project

1. **Purpose**

This memorandum documents a readiness assessment for beta testing of SendCUIEmail v0.5.2 with users who have limited technical capability but knowledge of CUI handling requirements.

2. **Target Audience Profile**

Beta Tester Characteristics:

- Familiar with CUI marking requirements (32 CFR Part 2002)
- Understands need for encryption when transmitting CUI via email
- Limited command-line experience
- Comfortable with Windows desktop applications
- May not know what “PowerShell” is

3. **Readiness Assessment**

Sender Experience (Encryption)

Feature	Status	Notes
Windows entry point	✓	Double-click <code>Encrypt.bat</code>
CUI category selection	✓	Clear menu with multi-select support
Password prompts	✓	Requirements shown upfront
Error messages	✓	Colored, clear guidance
Compliance reminders	✓	[COMPLIANT]/[NOT COMPLIANT] labels
Cleanup guidance	✓	.msg deletion reminder
Output location clarity	⚠	Folder created in current directory
Outlook integration	✓	.msg drafts auto-generated (Windows)

Recipient Experience (Decryption)

Feature	Status	Notes
README instructions	✓	Step-by-step included with files
Decrypt.bat available	✓	Double-click entry point
PowerShell one-liner	✓	Copy-paste decryption option
No external dependencies	✓	Uses built-in Windows APIs
File path handling	△	Requires understanding of file locations
Password visibility	△	No characters shown while typing

Documentation

Feature	Status	Notes
In-tool guidance	✓	Console shows compliance info
README for recipients	✓	Generated with each encryption
Quick-start guide	×	Not yet created
Video walkthrough	×	Not yet created

4. Identified Concerns**High Priority:**

- (a) **Output Location Confusion** – Files are created in a timestamped subdirectory of the current working directory. Non-technical users may not understand where “current directory” is.
- (b) **Password Entry UX** – Console password prompts show no characters while typing (security feature). Users may think nothing is happening.
- (c) **Recipient File Path Handling** – Recipients need to navigate to where they saved the .Locked files. No drag-and-drop support.

Medium Priority:

- (a) **No Quick-Start Guide** – Users would benefit from a 1-page PDF with screenshots.
- (b) **“What is PowerShell?”** – Some users have never seen a PowerShell window.
- (c) **Weak Password Acceptance** – Tool accepts “12345678” (meets 8-char minimum but is trivially guessable).

5. Assessment Summary

CONDITIONALLY READY FOR BETA

Rationale: Core functionality is complete and tested. The tool correctly encrypts files using FIPS-compliant algorithms, generates proper CUI markings, and provides compliance guidance. However, some UX friction exists for non-technical users.

Recommended for Beta With:

- Brief verbal walkthrough or screen share for first-time users
- Technical support contact available during initial testing
- Feedback collection on specific pain points

Not Recommended Without:

- Some form of user orientation (even informal)
- Expectation setting that this is command-line based

6. **Recommendations Before Wider Release**

- (a) Create a 1-page Quick-Start Guide (PDF with screenshots)
- (b) Add note about password entry: “Characters will not appear as you type”
- (c) Consider adding output path to final summary (full path, not just folder name)
- (d) Document the “where are my files?” answer prominently

7. **Implementation Plan (v0.5.3)**

The following improvements will be implemented before developer beta testing:

Priority 1: Password Entry Hint

Problem: SecureString input shows no characters; users think nothing is happening.

Solution: Add hint line before each password prompt:

(Characters will not appear as you type)
Enter encryption password:

Files Modified:

- `Encrypt.ps1` – Get-SecurePassword function (2 prompts)
- `Decrypt.ps1` – Get-Password function (1 prompt)

Priority 2: Output Location Clarity

Problem: Output folder shown in gray at bottom, easy to miss.

Solution: Move output folder to top of file list with yellow label and full paths:

Output folder:

C:\Users\Dev\Documents\SendCUIEmail_20260121_143022

Files ready to send:

C:\Users\Dev\Documents\SendCUIEmail_20260121_143022\Document.Locked

C:\Users\Dev\Documents\SendCUIEmail_20260121_143022\README.md

Files Modified:

- Encrypt.ps1 – Final summary section
- Decrypt.ps1 – Final summary section

Deferred Items

- **Weak Password Warning** – Deferred for developer beta; add if feedback indicates need
- **Quick-Start Guide PDF** – Deferred until non-technical user release

8. Beta Success Criteria

The following criteria define a successful developer beta test:

Required (Must Pass)

#	Criterion	Pass
1	Encrypt.bat launches without errors on Windows	☐
2	Files encrypt successfully with chosen password	☐
3	.Locked files and README.md are created in output folder	☐
4	Outlook .msg drafts are generated (CUI_Email and Password_Email)	☐
5	CUI banner marking appears correctly in email draft	☐
6	Recipient can decrypt using README.md instructions	☐
7	Decrypted files match originals (integrity verified)	☐
8	Wrong password correctly rejected with clear error	☐

UX Validation

#	Criterion	Pass
9	Password hint “(Characters will not appear...)” is visible	☐
10	Output folder location is clear and easy to find	☐
11	CUI category selection is understandable	☐
12	Compliance guidance (COMPLIANT/NOT COMPLIANT) is clear	☐
13	No confusing or unclear prompts encountered	☐

Success Threshold

- **Pass:** All 8 Required criteria pass, and at least 4 of 5 UX criteria pass
- **Conditional Pass:** All Required pass, but UX feedback requires minor changes
- **Fail:** Any Required criterion fails

Post-Beta Actions**If Pass:**

- Proceed to non-technical user preparation
- Create Quick-Start Guide PDF
- Consider v1.0.0 release candidate

If Conditional Pass:

- Address UX feedback in patch release (v0.5.4)
- Re-test affected areas
- Then proceed to non-technical preparation

If Fail:

- Document failure details
- Fix blocking issues
- Re-run beta test

9. Version Information

Version Assessed: v0.5.2
Assessment Date: January 21, 2026
Git Commit: 9f87a00
Implementation Target: v0.5.3